

STATISTICS

PART 1: Basic Concepts, Graphical Projection of Data, Frequency Distributions

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Basic Concepts of Statistics

Statistics

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Engineering, scientific, industrial, etc. problems can be solved through

Analytical Methods

(Exact or approximate solutions)

Numerical Methods

(Approximate solutions)

Statistical Methods or Artificial Intelligence Methods

(Estimated solutions)

Deterministic Approach

The problem is investigated under specific conditions.

Stochastic Approach

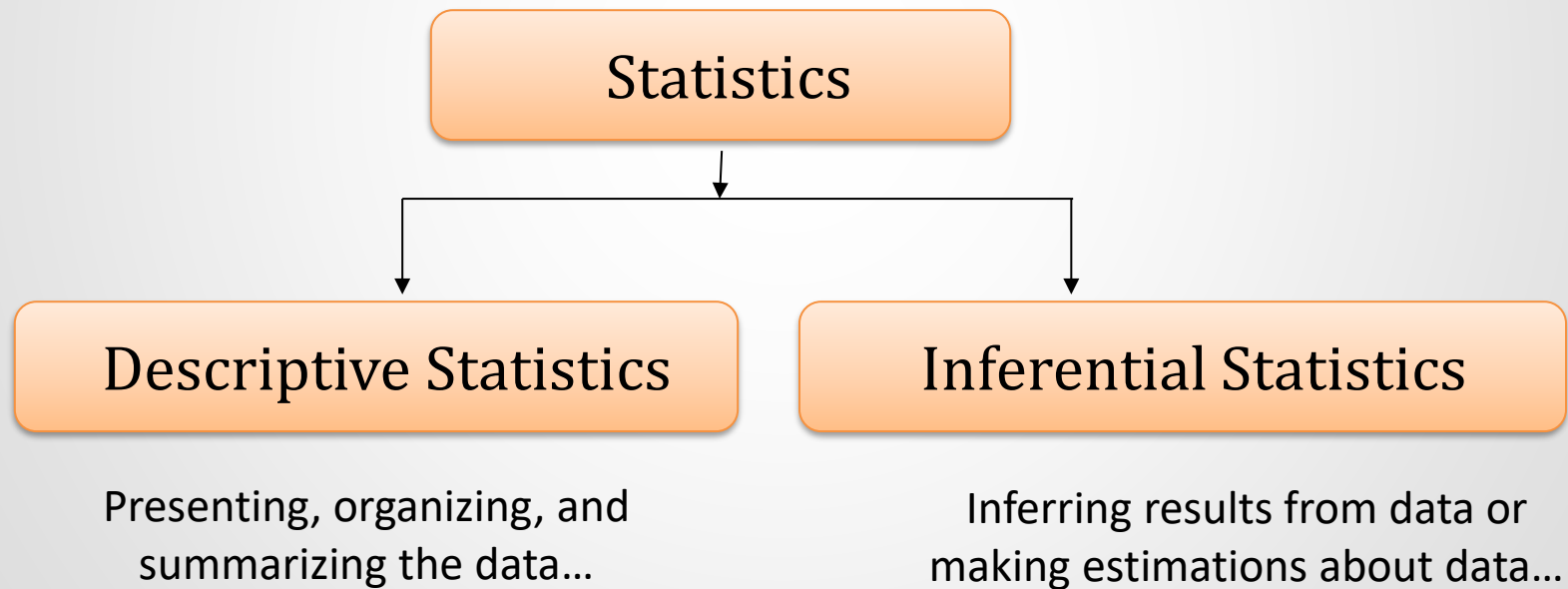
The problem is investigated under probabilistic conditions by using collected data.

Basic Concepts of Statistics

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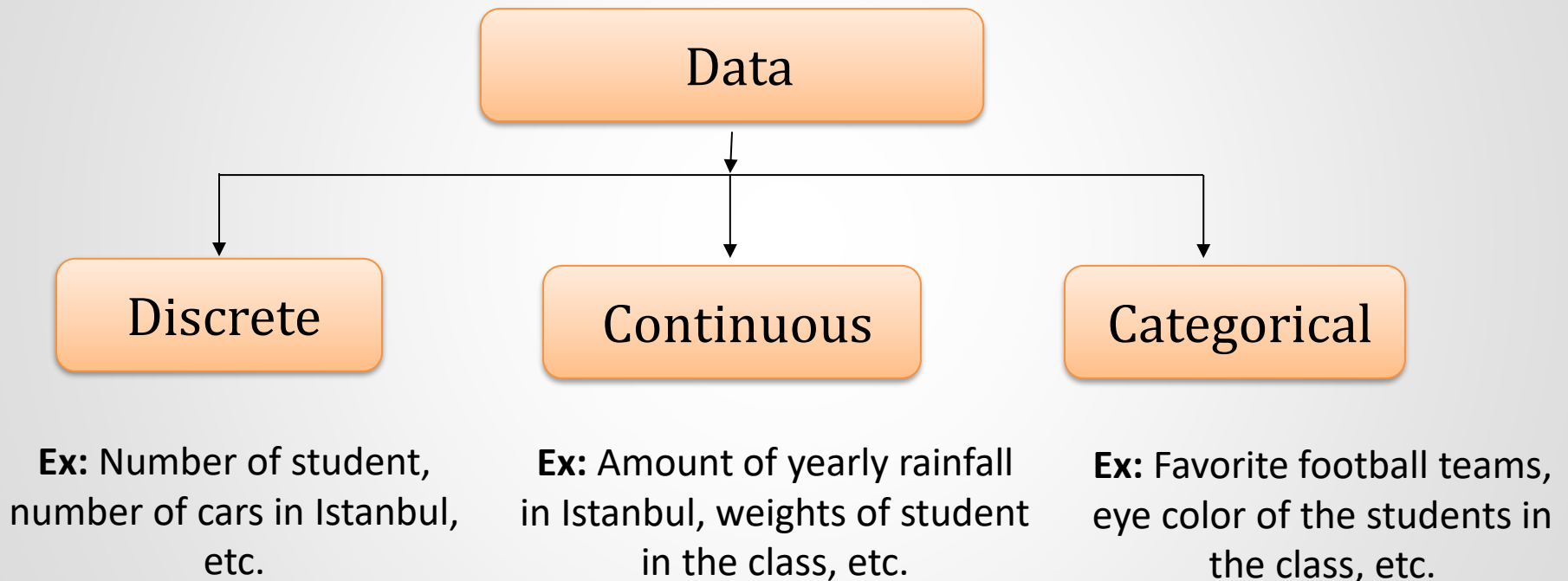
Definition: **Statistics** is the science of collection, organization, interpretation, and estimation of data.



Basic Concepts of Statistics

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Graphical Projection of Data

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Consider the following illustrative example...

EXAMPLE 1 (PUZZLE EXAMPLE): In order to learn in how many minutes a newly designed puzzle can be solved, 50 children were observed when they solved this puzzle. The following table shows solving times (in minutes) for the puzzle:

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19

Graphical Projection of Data

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How can we describe the given data?

4	15	6	11	2	15	9	21	13	10
17	19	11	8	11	13	12	8	17	9
15	24	14	7	13	16	14	12	15	14
5	10	13	23	12	31	13	25	5	16
11	15	26	17	16	14	28	10	15	19



The simplest organization of the data is sorting them.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Graphical Projection of Data

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The Sorted Data

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Min. value

$n = 50$

Max. value

$$\begin{aligned}\text{Range of the data} &= \text{Max. Value} - \text{Min value} \\ &= 31 - 2 \\ &= 29\end{aligned}$$

Graphical Projection of Data

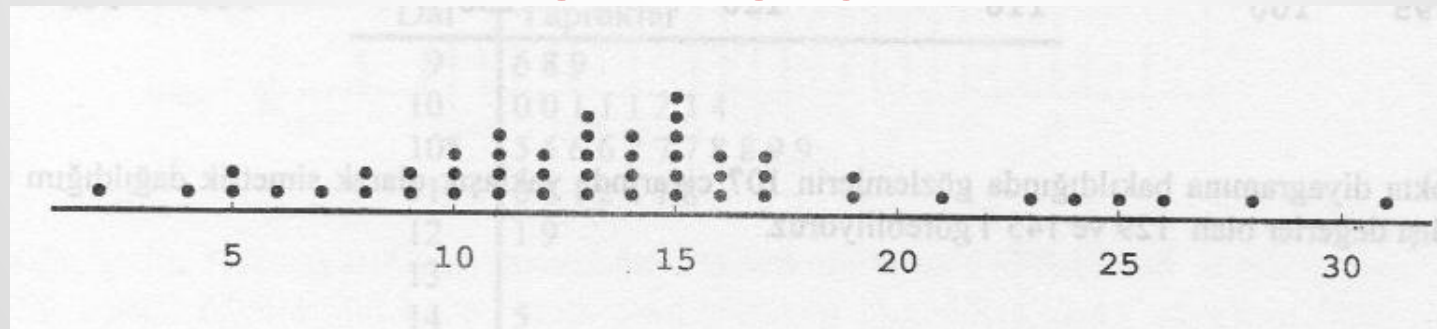
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2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

There are many different ways to illustrate data.

DOT PLOTS



Dot plots can be used to see the distribution of data, where data are concentrated or extreme values, etc.

Graphical Projection of Data

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2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

STEM-AND-LEAF PLOTS

Stem	Leaf
0	2 4 5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4 5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4 5 6 8
3	1

Since the given data set consists of one and two-digit numbers, here stem represents the tens-digit of the values while the leaf represents the ones-digit of the values.

Graphical Projection of Data

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2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

In order to obtain more illustrative stem-and-leaf plots, we can increase the number of stems.

Stem	Leaf
0	2 4
0*	5 5 6 7 8 8 9 9
1	0 0 0 1 1 1 1 2 2 2 3 3 3 3 3 4 4 4 4
1*	5 5 5 5 5 5 6 6 6 7 7 7 9 9
2	1 3 4
2*	5 6 8
3	1

How can stem-and-leaf plots be constructed for the data with non-integer values?

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2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

FREQUENCY DISTRIBUTION

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

Another way of showing the frequency of values in a data set is using a **frequency distribution (FD) table**.

$$\sum_i f_i = n = 50$$

Graphical Projection of Data

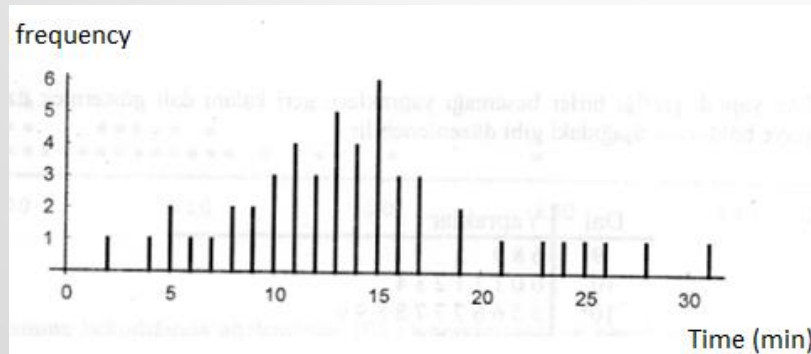
Statistics

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LINE PLOT

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

For the graphical display of frequency distribution tables, we often use **line plots**.



Graphical Projection of Data

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CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

When a data set contains continuous data or a large number of discrete data, it may not be appropriate to use **FD**. For such cases, we can use **classified frequency distribution (CFD)**.

Steps for Creating CFD:

1. Compute the range of data.
2. Determine the number of classes (k). In practice, k can be chosen generally between 5-15. You may use the following formulas to specify it:

$$k = \sqrt{n}$$
$$k = 1 + 3.3 \log n$$
$$k = 5 \log n$$

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CLASSIFIED FREQUENCY DISTRIBUTION AND HISTOGRAMS

3. Divide the data into classes with equal intervals. To create these classes, first compute the class interval:

$$\text{Class interval} = \left[\frac{\text{Range of data}}{\text{Number of classes } (k)} \right]$$

4. Find the midpoints (x_i) of each class.
5. Compute the frequency (f_i) of each class.

Graphical Projection of Data

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Ex: Create a classified frequency distribution table for the puzzle example.

2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

(1) Range = 29

(2) For the given data set, we choose k as 6. (Here we have not used the given formulas.)

$$(3) \text{ Class interval} = \left\lceil \frac{29}{6} \right\rceil = 5$$

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2	4	5	5	6	7	8	8	9	9
10	10	10	11	11	11	11	12	12	12
13	13	13	13	13	14	14	14	14	15
15	15	15	15	15	16	16	16	17	17
17	19	19	21	23	24	25	26	28	31

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2
(4)	(5)	$\sum_{i=1}^k f_i = 50$

Attention: In this course, we will assume that the value at the lower bound of a class belongs to that class, while the value at the upper bound does not.

For example, the value '7' belongs to the second class (7-12).

Graphical Projection of Data

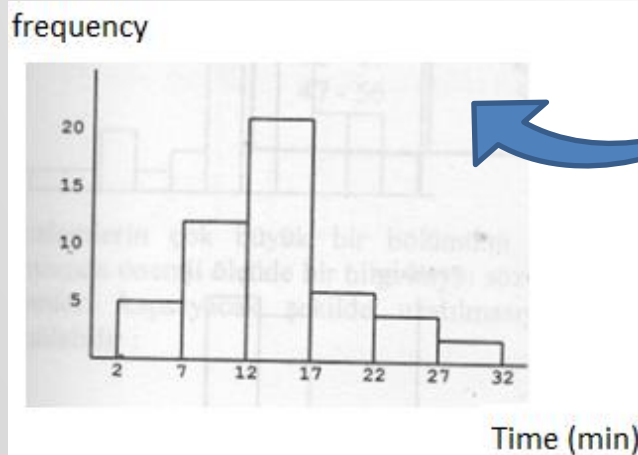
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HISTOGRAMS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2

For the graphical display of CFD tables, we use **histograms** instead of line plots.



- ❖ Here, rectangles represent the classes or mid-points.
- ❖ Histograms are not same with bar graphs!
- ❖ The area of each rectangle is proportional to the corresponding frequency.

Graphical Projection of Data

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CLASSES WITH UNEQUAL INTERVALS

Assume that 4 new data points are added to the data set: 28, 34, 41 and 55.

In this case;

$$\text{Range} = 55 - 2 = 53$$

If we select k as 6 again,

$$\text{Class interval} = \left\lceil \frac{53}{6} \right\rceil = 9$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-11	6,5	13
11-20	15,5	30
20-29	24,5	7
29-38	33,5	2
38-47	42,5	1
47-56	51,5	1
		Sum : 54

As seen from the CFD table, most of the data are concentrated in the second class. This results in a significant loss of information, specifically the details of data distribution). The issue here arises due to extreme values such as 41 and 55.

Graphical Projection of Data

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CLASSES WITH UNEQUAL INTERVALS

To overcome this problem, we can choose class intervals as 5 again, then extend the interval of the last class such as given below:

Classes (C_i)	Frequency (f_i)
2-11	13
11-20	30
20-29	7
29-38	2
38-47	1
47-56	1

For classes with
equal interval

Classes (C_i)	Frequency (f_i)
2-7	5
7-12	12
12-17	21
17-22	6
22-27	4
27-57	6

For classes with
unequal interval

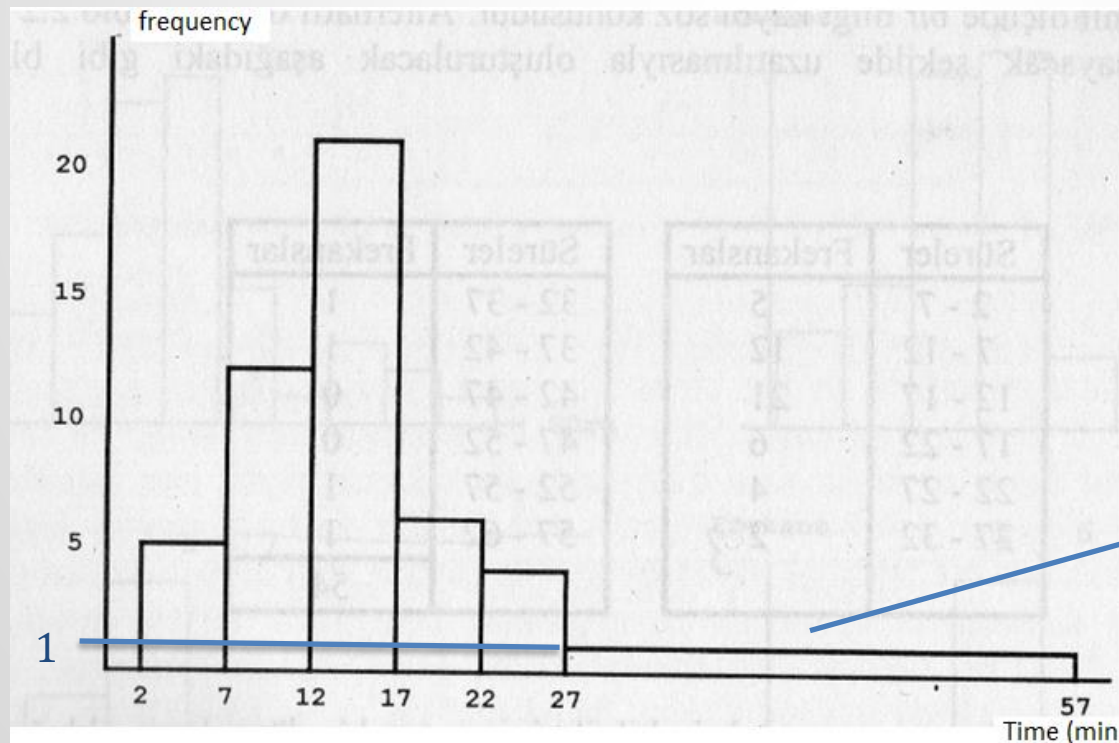
Graphical Projection of Data

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HISTOGRAM FOR CLASSES WITH UNEQUAL INTERVALS

The area of each rectangle in a histogram is equal to the corresponding frequency if the class intervals (width of rectangles) are normalized with 1. In this case, the sum of rectangle areas will be the number of data, i.e., n .



Why is the height of the rectangle 1 instead of 6?

Graphical Projection of Data

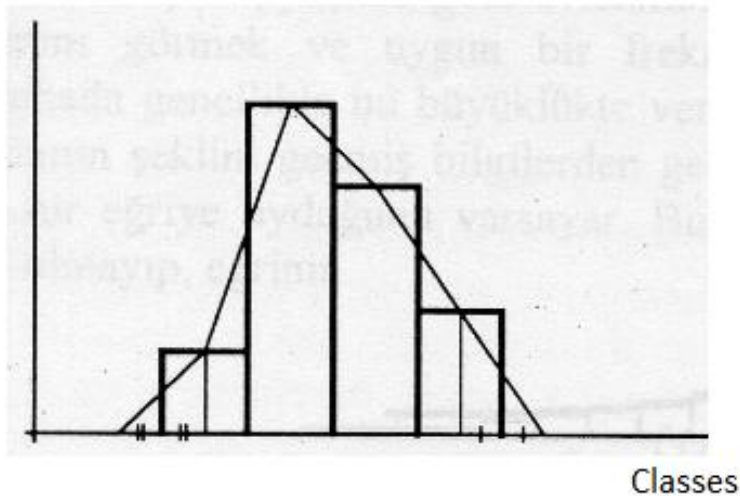
Statistics

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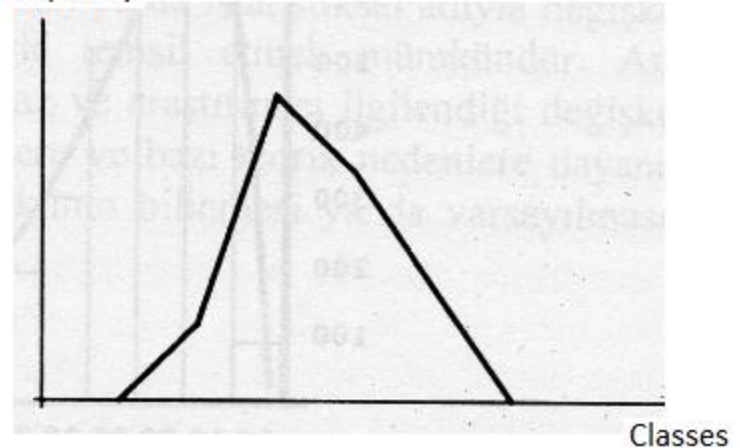
FREQUENCY POLYGON

A frequency polygon consists of the lines between midpoints of rectangles in a histogram. The first and the last line intersect the horizontal axis. The area below the frequency polygon is equal to the total area of rectangles.

Frequency



Frequency



Frequency Polygon

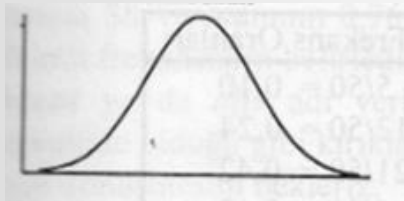
Graphical Projection of Data

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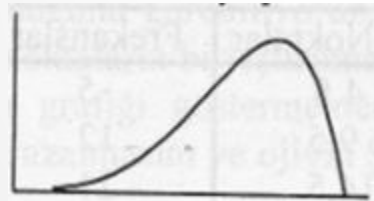
FREQUENCY CURVE

When the number of data (accordingly the number of classes) increases, the frequency polygon gets smoother. In this case, this polygon is called a **frequency curve**.

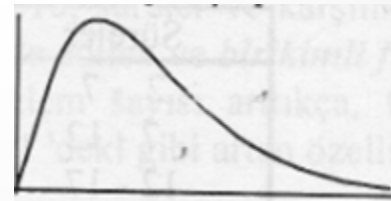
Some Basic Types of Frequency Curves



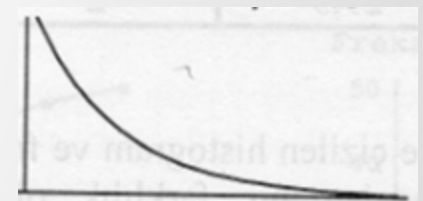
Symmetric



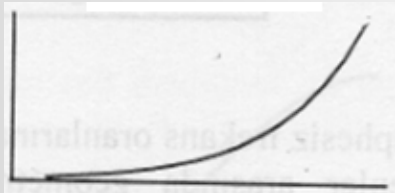
Skewed to left



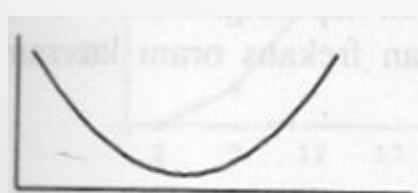
Skewed to right



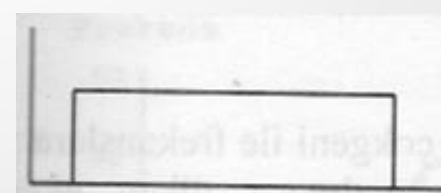
Inverse J-Shaped



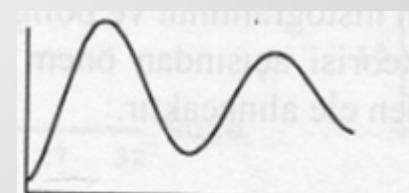
J-Shaped



U-Shaped



Rectangle



Two-mode

Graphical Projection of Data

Statistics

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RELATIVE FREQUENCY DISTRIBUTION

The **relative frequency** of a class $i, i=1,2,...,k$, can be computed as:

$$\text{Relative Frequency } (f_i^R) = \frac{\text{Frequency } (f_i)}{\text{Number of data } (n)}$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)
2-7	4,5	5	5/50=0,10
7-12	9,5	12	12/50=0,24
12-17	14,5	21	21/50=0,42
17-22	19,5	6	6/50=0,12
22-27	24,5	4	4/50=0,08
27-32	29,5	2	2/50=0,04

$$\sum_i f_i^R = 1$$

Note: For relative frequency, the area of the histogram is 1 if the width of each rectangle is 1. Accordingly, the area under the frequency polygon or curve is 1.

Discuss the relationship between this fact and probability functions!!!

Graphical Projection of Data

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(INCREASING) CUMULATIVE FREQUENCY DISTRIBUTIONS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Increasing) Cumulative Frequency ($F_i \nearrow$)	Cumulative Relative Frequency ($F_i^R \nearrow$)
2-7	4,5	5	$5/50=0,10$	5	0,10
7-12	9,5	12	$12/50=0,24$	17	0,34
12-17	14,5	21	$21/50=0,42$	38	0,76
17-22	19,5	6	$6/50=0,12$	44	0,88
22-27	24,5	4	$4/50=0,08$	48	0,96
27-32	29,5	2	$2/50=0,04$	50	1,00

Graphical Projection of Data

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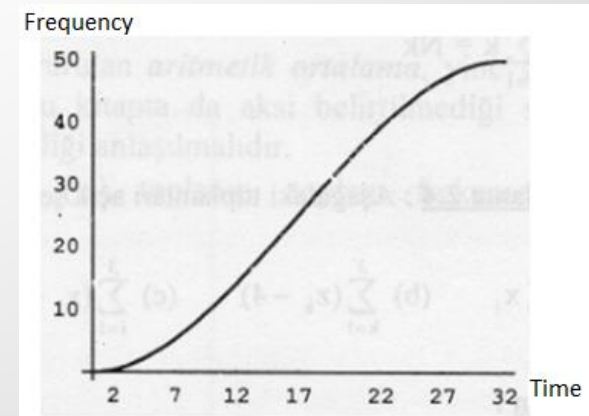
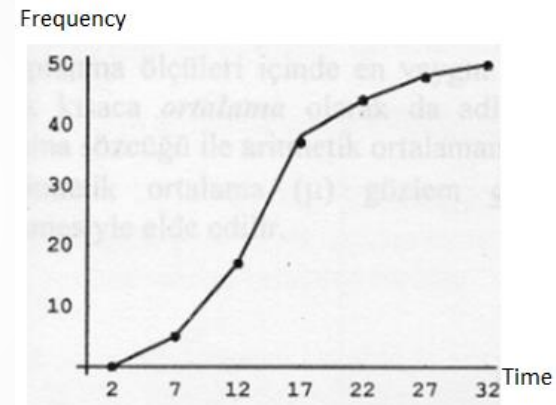
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CUMULATIVE FREQUENCY POLYGON AND CURVE

Classes (C_i)	(Increasing) Cumulative Frequency ($F_i \nearrow$)
2-7	5
7-12	17
12-17	38
17-22	44
22-27	48
27-32	50

Cumulative
Freq. Polygon

Cumulative
Freq. Curve



Graphical Projection of Data

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(DECREASING) CUMULATIVE FREQUENCY DISTRIBUTION

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Decreasing) Cumulative Frequency ($F_i^{\searrow}$)	Cumulative Relative Frequency ($F_i^{R\searrow}$)
2-7	4,5	5	$5/50=0,10$	50	1,00
7-12	9,5	12	$12/50=0,24$	45	0,90
12-17	14,5	21	$21/50=0,42$	33	0,66
17-22	19,5	6	$6/50=0,12$	12	0,24
22-27	24,5	4	$4/50=0,08$	6	0,12
27-32	29,5	2	$2/50=0,04$	2	0,04

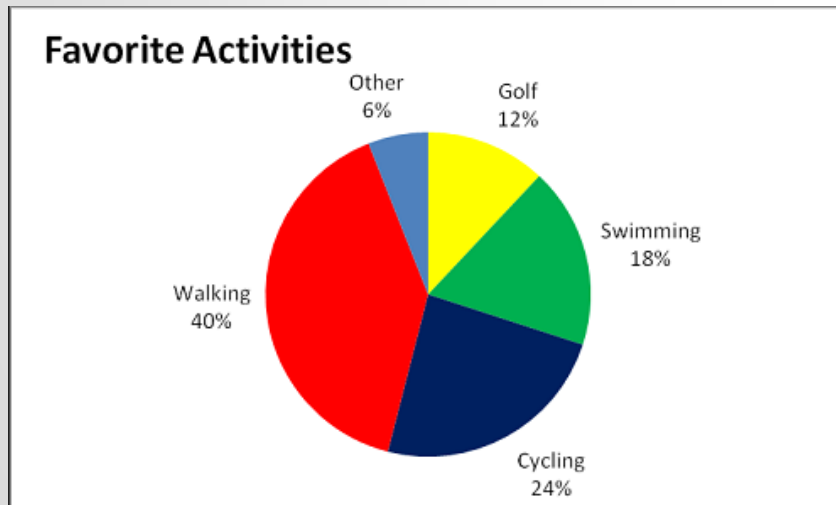
Graphical Projection of Data

Statistics

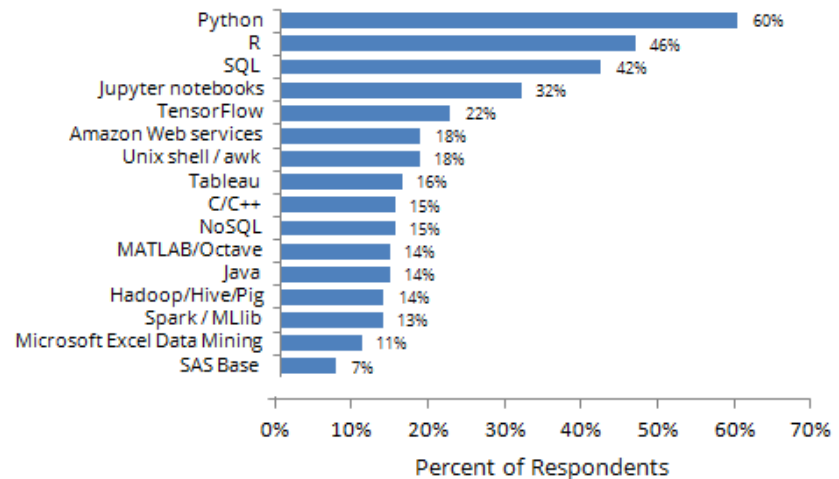
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For the projection of numerical data, we investigated dot plots, line plots, histograms, frequency polygons, etc. When data are categorical, we can use many figures such as circular (pie) charts and bar charts.

Data science tools, technologies, languages used in 2017



A Circular Chart (1)



A Bar Chart (2)